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Fas and FasL in the homeostatic regulation of immune responses

David H. Lynch, Fred Ramsdell and Mark R. Alderson

Studies of the biological effects of Fas signaling, using transformed cell lines as targets, indicate that ligation of the Fas receptor induces an apoptotic death signal. Chronically activated normal human T cells are also susceptible to Fas-mediated apoptosis. However, interactions between Fas and Fas ligand can also yield a costimulatory signal. Here, David Lynch, Fred Ramsdell and Mark Alderson present a model for the role of Fas and FasL in the homeostatic regulation of normal immune responses. They discuss how dysregulation of the Fas apoptotic pathway may contribute to certain disease states, including autoimmune disease and human immunodeficiency virus (HIV)-induced depletion of CD4$^+$ T cells.

Fas (APO-1, CD95) was originally described as a cell-surface molecule that could mediate apoptotic cell death of transformed cells[1,2]. In addition, chronically activated normal human T cells (e.g. T-cell clones) are also susceptible to Fas-mediated apoptosis[3]. The recent demonstration that the lymphoproliferative diseases in *lpr/lpr* and *gld/gld* mice can be attributed to defects in the genes encoding Fas and Fas ligand (FasL), respectively[4–6], indicates an important role for these proteins in regulating normal immune responses and maintaining self-tolerance. Interestingly, stimulation through Fas does not always induce a 'death signal'. Indeed, ligation of Fas on freshly isolated T cells has been demonstrated to costimulate cellular activation and proliferation[7]. Thus, it seems clear that Fas can mediate opposite effects depending on the state of activation of the T cells. The dramatic differences in the outcome of Fas-mediated signal transduction appear to be a function of the activation state of the responding cells. On the basis of these results, a general model may be constructed in which signaling through Fas in the early stages of an immune response augments the generation of an effective immune response, whereas signaling through Fas in the later stages leads to downsizing of an immune response at the clonal level via apoptosis.

Fas and T-cell activation

Fas and FasL are highly regulated in their expression. Whereas most cultured T cells constitutively express Fas, freshly isolated peripheral blood T cells can be broadly divided into two groups based on their expression of Fas: naive (CD45RA$^+$) T cells express little or no cell-surface Fas (although it is rapidly induced upon activation) and previously activated (CD45R0$^+$) T cells express relatively high amounts of cell-surface Fas (Refs 8,9). Under normal conditions, neither CD45RA$^+$ nor CD45R0$^+$ T cells appear to express FasL constitutively; however, FasL is rapidly expressed following T-cell activation. Despite the fact that freshly isolated peripheral blood T cells express (or can be induced to express) Fas, these cells are not sensitive to Fas-mediated apoptosis. Instead, ligation of Fas on freshly isolated T cells by monoclonal antibody (mAb)[7] or recombinant FasL (M.R. Alderson, F. Ramsdell and D.H. Lynch, unpublished), in the presence of suboptimal stimulation via the T-cell receptor (TCR), leads to enhanced proliferation, increased expression of CD25 and CD69, and increased production of cytokines such as interleukin 2 (IL-2), interferon γ (IFN-γ) and tumor necrosis factor (TNF).

On the basis of these observations, a model is proposed here that suggests Fas-mediated signaling may

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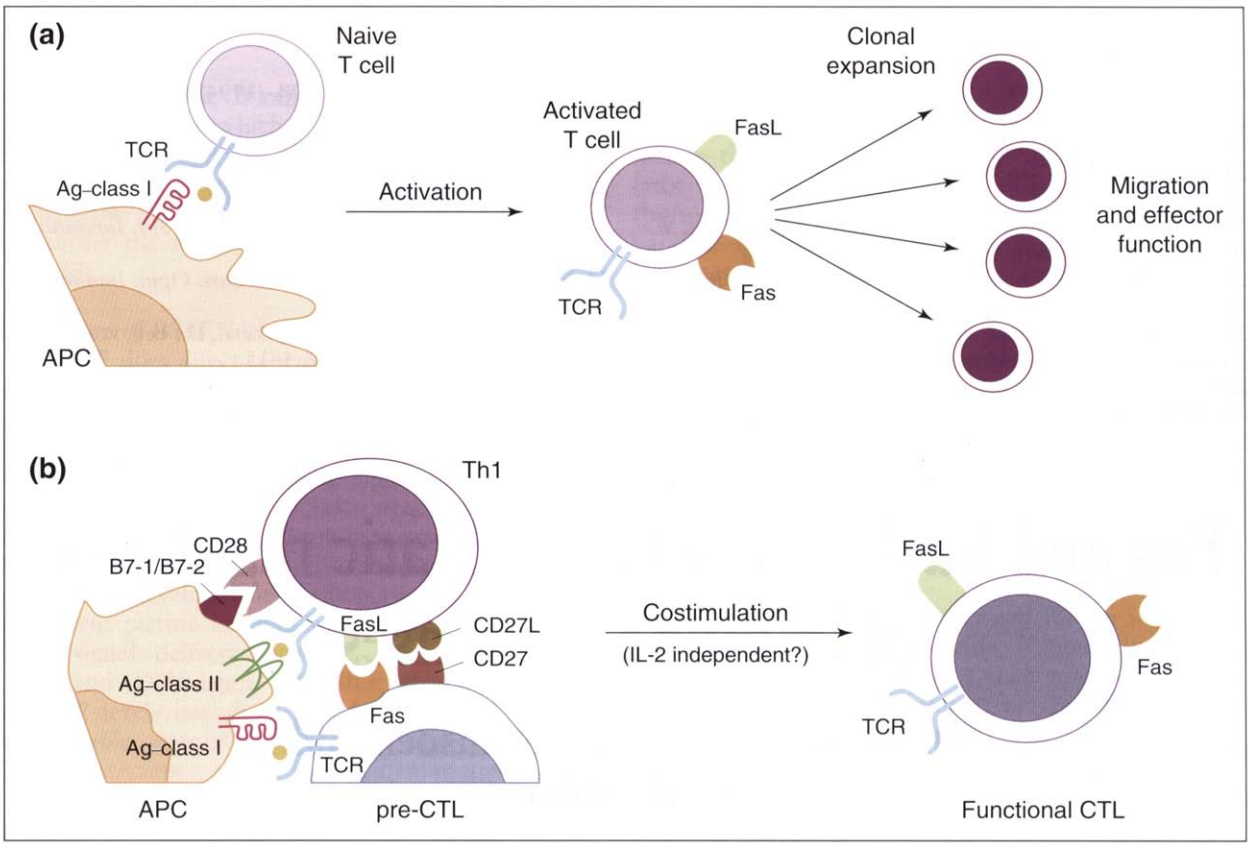

Fig. 1. *Functions of Fas and FasL in the costimulation of T-cell function. (a) Stimulation of the TCR by peptide antigens presented by either MHC class I (to CD8⁺ T cells) or MHC class II (to CD4⁺ T cells) results in expression of Fas and FasL. This leads to enhanced clonal expansion and effector function. (b) Although the precise role of other cell-surface molecules in Fas-mediated costimulation is unclear, the interaction between FasL on a Th1 cell and Fas on a pre-CTL may contribute to differentiation of the latter into a functional CTL, potentially in an IL-2-independent manner. Abbreviations: Ag, antigen; APC, antigen-presenting cell; CTL, cytotoxic T lymphocyte; FasL, Fas ligand; IL-2, interleukin 2; MHC, major histocompatibility complex; TCR, T-cell receptor; Th1, T helper 1.*

contribute to the generation of an immune response. Figure 1 depicts the activation of a naive $CD4^+$ T cell by an antigen-presenting cell (APC), resulting in the expression of Fas and FasL. The precise role of other cell-surface molecules such as CD28, CTLA-4, and their counter-structures, B7-1 and B7-2, in Fas-mediated costimulation is unclear at present. However, the interactions between FasL expressed by the $CD4^+$ T helper (Th) cell and the Fas receptor on another lymphoid cell [such as the pre-cytotoxic T lymphocyte (CTL) shown in Fig. 1, which is also being stimulated through its TCR] may contribute to the differentiation of the pre-CTL into a functional CTL. Fas-mediated costimulation appears to be IL-2 independent[7]; thus, secretion of cytokines such as IL-2 may not be required for activation of the pre-CTL. As an immune response progresses, however, the role of Fas and FasL is altered. As noted previously, there are qualitative differences in how cells perceive signals through Fas, depending, partly, on the differentiation state of the cell and possibly also on the cell cycle.

Fas and activation-induced cell death (AICD)

Although Fas–FasL interactions may play a costimulatory role in the generation of an immune response, there is a clear role for these molecules in immune regulation. When the TCR/CD3 complex of primary T cells is engaged either by APCs presenting antigenic peptide or by antibody to TCR/CD3, these cells normally undergo a series of activation events that includes cell proliferation and cytokine production. By contrast, stimulation of previously activated T cells through the TCR/CD3 complex induces them to undergo apoptotic cell death, a process that has been termed activation-induced cell death (AICD)[10]. It has been proposed that AICD serves two functions *in vivo*: first, to limit the expansion of an immune response by eliminating lymphocytes that are no longer needed; and, second, to eliminate autoreactive T cells that have escaped thymic selection.

The fact that previously stimulated T cells from either *lpr/lpr* or *gld/gld* mice do not undergo AICD implicates Fas–FasL in this process[11–13]. It is important to know whether AICD in human T-cell clones and T-cell lines is also mediated by Fas–FasL interactions and whether this process can be blocked by a Fas-specific mAb that inhibits target cell apoptosis induced by FasL (Ref. 14). The data demonstrate that chronically activated human T cells constitutively express high amounts of Fas and are susceptible to Fas-mediated killing, either by agonistic Fas mAbs or by recombinant FasL. Ligation of the TCR/CD3 complex on these T cells rapidly induces expression of FasL and ultimately leads to cell death[15]. Antagonists of Fas, such as soluble Fas M3 mAb and the Fas–Fc fusion protein, are able to block AICD induced by CD3 antibody, or by phorbol ester plus calcium ionophore, or by staphylococcal enterotoxin B (SEB) superantigen[15]. Recent studies by

three other groups also indicate that Fas–FasL interactions play a dominant role in mediating AICD in human and murine T cells[16–18].

Although previously activated T cells are susceptible to Fas-mediated apoptosis, *in vitro* stimulation induces AICD in only a proportion of these cells. Thus, the net result is clonal expansion[19]. These results suggest that APCs provide a critical costimulatory signal that can overcome the mediators of AICD in at least a proportion of the T cells. The nature of this APC-derived signal is unclear but does not appear to involve either CD28 or IL-2 (Refs 20,21).

Why Fas transduces a costimulatory signal in resting T cells but an apoptotic signal in previously activated T cells is unclear at present. There is some evidence that expression of Bcl-2 (Ref. 22), or other members of the Bcl-2 family, may result in the resistance or sensitivity of T cells to apoptosis[23–25]. Alternatively, the newly identified Fas-associated phosphatase, FAP-1, which inhibits Fas signal transduction[26], may contribute to Fas-mediated activation instead of cell death.

Although recent *in vitro* studies on Fas expression and function on human thymocytes have suggested a role for Fas in negative selection[27,28], analysis of *lpr/lpr* mice has indicated that the Fas receptor is not critical for TCR repertoire selection in the thymus[29–32]. However, evidence has accumulated to suggest that the Fas-mediated AICD pathway may function as a mechanism by which autoreactive T cells that have escaped thymic negative selection can be eliminated in the periphery. Indeed, the lymphadenopathy and autoimmune pathology associated with defects in Fas/FasL function in *lpr/lpr* and *gld/gld* mice suggest a critical role for this pathway in maintaining peripheral tolerance *in vivo*. A recent study that analyzed TCR-transgenic mice crossed with MRL-*lpr/lpr* mice also indicated that Fas plays a critical role in AICD in mature T cells but does not play a role in negative selection in the thymus[33]. However, it seems likely that factors in addition to Fas are also involved in T-cell elimination *in vivo*. For example, in Fas-deficient *lpr/lpr* mice, the deletion of Vβ8 T cells following injection of SEB is impaired but not totally absent (Ref. 34; F. Ramsdell, unpublished). In this regard, recent studies have suggested that interactions between other cell-surface receptor–ligand pairs [e.g. intercellular adhesion molecule 1 (ICAM-1), leukocyte function-associated molecule 1 (LFA-1) (Ref. 20) and 4-1BB/4-1BBL (Ref. 35)] and other members of the TNF family, including TNF (Refs 36,37), CD30L and CD40L (M.R. Alderson, unpublished), may affect (either by enhancing or inhibiting) Fas-mediated AICD.

Expression of FasL on Th-cell subsets

FasL was originally identified on a $CD8^+$ T-cell hybridoma that induced target cell lysis in a Ca^{2+}-independent, Fas-dependent manner[38]. Although FasL does not appear to be the primary cytotoxic molecule associated with the induction of target cell lysis, a variety of studies have indicated that FasL can act as a second, perforin-independent mechanism by which $CD8^+$ T cells induce target cell cytolysis[39–41]. The ability of $CD4^+$ T cells to act as cytolytic effector cells in the absence of perforin-containing granules suggests that these cells also use the Fas pathway[42,43].

The expression of Fas and FasL has recently been evaluated on Th-cell subsets, focusing on expression of these molecules by Th1 and Th2 cells. The data indicate that, whereas Fas is expressed by both Th subsets, only Th1 cells express readily detectable amounts of FasL after activation[21]. A recent study confirmed expression of FasL by activated Th1 cells, but suggested that a subset of Th2 cells also expresses FasL (Ref. 44). Data derived from short-term Th-cell lines also indicate that FasL is expressed predominantly, but not exclusively, by Th1 cells[21]. Interestingly, the differential expression of FasL by Th1 and Th2 cells correlates with the ability of these cells to undergo AICD. Whereas Th1 clones are highly sensitive to AICD, Th2 clones are relatively resistant. However, the Th2 clones do express Fas because these cells are sensitive to the apoptotic effects of FasL expressed by Th1 cells. Human Th1 cells, but not Th2 cells, also induce cytolysis in certain target cells[45]; thus, it is likely that FasL expression in human Th cells is also predominantly confined to the Th1 subgroup.

AICD in clonal downsizing

The second part of the model describing the role of Fas and FasL in the regulation of normal immune responses concerns the role of AICD in clonal downsizing. Figure 2 depicts a mixture of antigen-specific T cells [such as $CD8^+$ or $CD4^+$ cells, which mediate CTL or delayed-type hypersensitivity (DTH), respectively] engaged in the recognition and elimination of their cognate target cells (such as tumor cells, virally infected somatic cells or hapten-modified cells). The consequences of continuous exposure of T cells to antigen include the induction of a state of susceptibility of the T cells to Fas-mediated apoptosis[3,9] and the persistent expression of FasL. Early in the course of the effector cell response, when the ratio of effector cells to target cells is relatively low, the probability of T–T-cell interactions is also likely to be relatively low. Thus, although the T cells are expressing Fas and FasL, little or no AICD of the T cells is apparent. However, as the target cells are eliminated, the ratio of effector cells to target cells undoubtedly increases dramatically, and the opportunity for activated T cells expressing both Fas and FasL to interact also increases. The result of such interactions is the apoptotic death of a proportion (perhaps a majority) of the activated T cells in the local area. It is clear that T cells can die by both suicide[16,17] and fratricide[46] *in vitro*; however, determination of which of these two mechanisms of cell death is important *in vivo* is unclear.

Why, then, do all of the activated T cells not undergo Fas-mediated death in this scenario? The answer may lie in the fact that expression of FasL is tightly regulated and requires continuous stimulation. Thus, as the level of antigenic stimulation of the T cells decreases (due to elimination of the target cells), the number of cells expressing FasL also decreases. As a result, although a proportion of the T cells is eliminated via AICD, a proportion of the cells also survives. Perhaps it is these cells that are destined to become memory T cells.

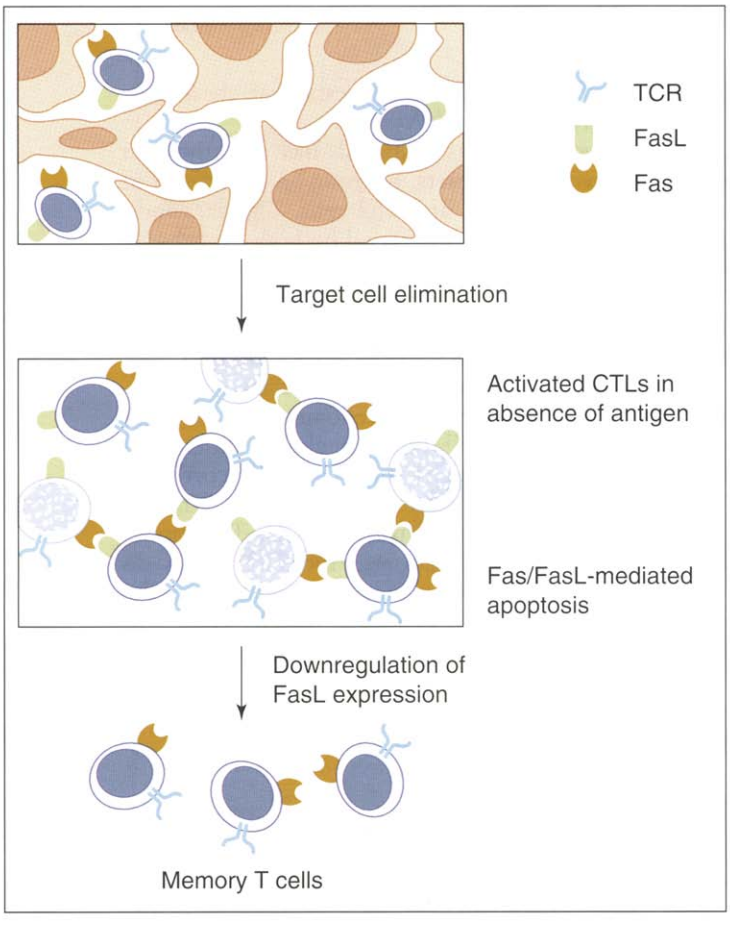

Fig. 2. *Functions of Fas and FasL in clonal downsizing of an immune response via AICD. Recognition of target cells by antigen-specific T cells results in target cell elimination and continuous stimulation of the effector T cells which, in turn, induces expression of Fas and FasL. As target cells are eliminated, the likelihood of T–T-cell interactions increases, thereby promoting clonal downsizing via Fas-mediated AICD. Elimination of cells expressing the target antigen also eliminates the stimulatory signal for the T cells, leading to downregulation of FasL expression. The proportion of T cells that escapes AICD go on to form memory cells. Abbreviations: AICD, activation-induced cell death; for other abbreviations see legend to Fig. 1.*

Potential pathological consequences of abnormal Fas–FasL interactions

The best support for the role of Fas and FasL in the clonal downsizing of immune responses is found in mice homozygous for either the *lpr* or *gld* mutations. As a result of these mutations, the Fas–FasL interaction pathway in AICD is rendered inoperative, and clonal downsizing of immune responses is dramatically impaired. As a consequence of the inability to eliminate lymphocytes that have been propagated in response to antigenic stimulation, the total number of lymphoid cells in an affected animal would be expected to increase progressively. The accumulation of lymphocytes observed in the periphery of *lpr/lpr* and *gld/gld* mice[47] is consistent with this prediction. Other aspects of the phenotype of *lpr/lpr* and *gld/gld* mice also suggest that the Fas–FasL pathway is involved in the maintenance of self-tolerance. In addition to the accumulation of lymphocytes, these mice develop a systemic lupus erythematosus (SLE)-like autoimmune disease characterized by hypergammaglobulinemia, the production of antibodies both to single-stranded and double-stranded DNA, and the induction of rheumatoid factor and glomerulonephritis (reviewed in Refs 47,48).

Whether dysregulation of Fas–FasL interactions plays an important role in the genesis of autoimmune disease in humans remains to be addressed. Although results from a recent study suggested that patients with SLE have elevated levels of soluble Fas (Ref. 49), other studies have failed to confirm this observation[50]. Interestingly, results from a very recent study of children with autoimmune lymphoproliferative syndrome, a disease with symptoms virtually identical to those manifest in *lpr/lpr* and *gld/gld* mice, contain dominant-negative mutations in the *fas* gene[61].

CD4 and Fas-mediated AICD

Just as an inability to transmit signals via the Fas–FasL receptor–ligand pair may lead to pathological conditions, so may overexpression and inappropriate signaling. In 1991, Ameisen and Capron proposed that the depletion of $CD4^+$ T cells in human immunodeficiency virus (HIV)-infected patients was a result of inappropriate induction of apoptosis[51]. Indeed, freshly isolated T cells from a relatively high proportion of HIV^+ individuals have been found to undergo AICD either spontaneously or when cultured with T-cell mitogens[52,53]. As noted earlier, AICD is not observed in freshly isolated T cells from healthy individuals, but is found in cultures of chronically activated T cells. Thus, the qualitatively similar AICD observed between chronically activated cells derived from normal donors and freshly isolated T cells from HIV-infected donors suggests the likelihood of a similar underlying mechanism.

Although circumstantial, supportive evidence of a role for FasL in T-cell depletion in HIV patients has been reported. Initial studies by Newell *et al.*[54] have demonstrated that crosslinking of CD4 on murine T cells primes these cells to undergo AICD. Similar results have been obtained with normal human T cells when CD4 is crosslinked by either the Leu3A mAb or gp120/anti-gp120 immune complexes[55,56]. Wang *et al.*[57] have confirmed and extended these results by demonstrating that normal mice, but not *lpr/lpr* mice, show depletion of $CD4^+$ T cells *in vivo* after treatment with an antibody to CD4. Thus, crosslinking of CD4 either primes or induces apoptotic death in these cells in a Fas-dependent manner. Finally, gp120/anti-gp120 complexes have also been demonstrated to induce deletion of T cells in mice expressing a human CD4 transgene[58]. Collectively, these observations suggest that crosslinking of CD4 on Th cells has two important effects: it provides a stimulatory signal necessary for expression of FasL and it induces susceptibility to Fas-mediated apoptosis. In this regard, CD4 crosslinking has been reported to upregulate Fas expression by T cells[59]. Thus, a model for T-cell depletion in HIV infection involves gp120/anti-gp120 immune complexes inducing inappropriate AICD in T cells through a Fas-dependent phenomenon.

The fact that Th1 cells can be induced to express FasL, whereas Th2 cells express little or no cell-surface FasL, also leads to an interesting possibility regarding the progressive elimination of $CD4^+$ cells in humans infected with HIV. Th1 and Th2 cells may both be stimulated by Ab-mediated crosslinking of gp120–CD4 complexes. Whereas both populations of T cells would be expected to become susceptible to Fas-mediated

apoptosis, only (or primarily) Th1 cells express FasL. Thus, the activated Th1 cells may be responsible for inducing AICD in both cell populations. A 'competitive advantage' (or perhaps, more properly, a 'competitive disadvantage') may exist for Th1 cells to interact among themselves rather than with Th2 cells. This could be due to the fact that the frequency of Th2 cells is lower than Th1 cells. In addition, stimulation would be expected to lead to clonal proliferation characterized by small clusters of activated Th1 cells that are susceptible to Fas-mediated apoptosis and express FasL. Such a situation could easily be envisaged as leading to the preferential loss of Th1 cells, resulting in a relative increase in the proportion of Th2 cells. Although still a matter of some controversy, an increase in the frequency of Th2 cells has been associated with the progression of disease in HIV-infected individuals (reviewed in Ref. 60).

Concluding remarks

Analysis of the *lpr* and *gld* molecular defects, and the immunological consequences of these defects, has provided considerable insight into the basis for the autoimmune disease process in strains of mice homozygous for these mutations. Results of more-recent work regarding the biological consequences of Fas–FasL interactions in different subpopulations of lymphocytes, and in cells of different activation states, may have profound implications regarding both the regulation of normal immune responses and the pathological conditions arising from either too little or too much interaction via this pathway. Many important questions remain to be answered. For example, what are the differences between naive and chronically activated T cells that lead to the profound difference in their responses to Fas ligation (i.e. costimulation versus apoptosis)? What other factors may regulate AICD? Is cell–cell contact absolutely required to induce AICD or can FasL function in normal biological settings as a soluble factor? What are the roles of other cell interaction molecules [e.g. CD28/CTLA-4 and B7-1/B7-2; ICAM-1 and LFA-1; and major histocompatibility complex (MHC) class I and class II molecules, TNF receptor (TNFR), CD30, 4-1BB, and their respective ligands] in altering Fas-mediated signaling? In addition to providing a platform from which to begin answering these questions, the model presented here may also be useful in understanding the role of these receptor–ligand pairs in the regulation of immune responses.

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Nitric oxide production by human monocytes: evidence for a role of CD23

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Nitric oxide (NO) appears to be an important and pleiotropic bioregulator of immune responses. The existence of the NO synthase (NOS) pathway in human monocytes/macrophages remains a subject of controversy, despite an increasing number of reports suggesting that human monocytes produce NO in vitro *in response to various stimuli. Here, Bernard Dugas and colleagues consider the arguments supporting these conclusions, with particular emphasis on the results obtained by ligation of the low-affinity IgE receptor (FcεRIIb/CD23b).*

Nitric oxide (NO) has recently gained attention as a new class of messenger molecule that serves a variety of functions in different tissues[1–3]. It is diffusible and activates biochemical processes inside the producer cells as well as within neighbouring cells. NO is the final product of the oxidation of guanidino nitrogen of L-arginine by nitric oxide synthase (NOS; EC 1.14.13.39).

Several NOS isoforms have been identified, representing the products of three separate genes on distinct chromosomes; these genes have been cloned in rodents and humans[4]. The isoforms vary considerably in their intracellular location, structure, regulation and functions. Two constitutive NOS (cNOS) isoforms, which are spontaneously expressed in various tissues, have been demonstrated: the neuronal cNOS (type I NOS) was first identified in the cytosol of central and peripheral neurons; the endothelial cNOS (type III NOS) is generally membrane bound and is found in endothelial cells. Following stimulation by transient calcium fluxes, cNOS isoforms generate small amounts of NO for short periods of time.

Another isoform of NOS, usually absent in resting cells, can be induced by a variety of stimuli, such as

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